Reflect-Aug-Seg

Reflectivity-Augmented LiDAR Scene Understanding
for Robotic Perception

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The Problem & Core Idea

The Problem

Raw LiDAR intensity != reflectivity

Intensity is entangled with range, incidence angle, and sensor response, treating it as a surface property is misleading.

Our Approach

A lightweight pseudo-reflectivity proxy

Multiply intensity by range to partially compensate for distance decay, cheap, interpretable, no calibration needed.

Key Question

Can a physically motivated, range-aware signal augmentation improve semantic interpretability and lightweight downstream usefulness in LiDAR scene understanding?

- **Signal-level analysis**
  How does I·R reshape intensity?
- **Temporal evaluation**
  Does it generalize across frames?
- **Downstream utility**
  Does a classifier benefit?

Mathematical Model

LiDAR Return-Strength Model

Each point: $p_i = (x, y, z, I)$

Range: $R = \sqrt{x^2 + y^2 + z^2}$

Intensity model:

$$I \propto \rho \cdot \cos(\alpha) / R^2$$

Pseudo-Reflectivity Proxy

$$\hat{\rho} = I \cdot R$$

Not calibrated reflectivity: a heuristic, range-aware transformation that exposes more semantic structure at minimal cost.

Proxy Family Evaluated

- I (raw baseline)
- $I \cdot R$ (direct proxy)
- **$I \cdot R^2$ (aggressive — rejected)**
- **$\log(1 + I \cdot R)$ (stabilized ✓)**
- Percentile-normalized $I \cdot R$
- Robust-scaled $I \cdot R$

Metric: Fisher separability $S = \sigma^2 B / (\sigma^2 W + \varepsilon)$

Pipeline: How We Did It

01
Baseline Lock
Midterm logic re-executed; dataset + proxy verified

02
Window Setup
15 windows: 5 short, 5 medium, 5 long over seq 00

03
Proxy Variants
Build & characterize I·R, $I{\cdot}R^2$, log(1+I·R), etc.

04
Global Signal
Per-window statistics across all 15 windows

05
Semantic Analysis
Fisher separability gains per window per proxy

06
Failure Analysis
Classify windows: strong / weak / failure

07
Visual Artifacts
600-frame BEV + FPV renderings, GIF, CSV

08
Downstream Test
Logistic regression: [I] vs [I, I·R]

Dataset: SemanticKITTI Sequence 00 • 4,541 frames • Modular Python codebase

Results & Demo

600-Frame Contiguous Run

+14.45% Relative gain in Fisher separability

400/600 Frames with positive gain

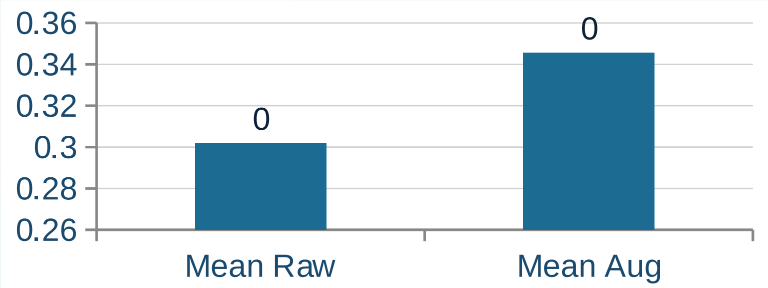
+0.0436 Mean delta (aug − raw)

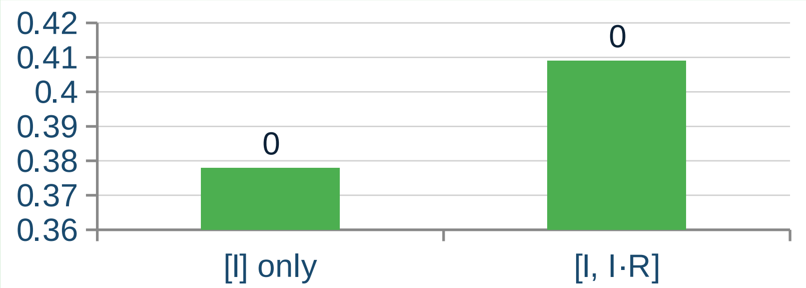
Multi-Window Analysis

`log(1 + I·R)` consistently positive across 14/15 windows
I·R shows strong gains but has explicit failure modes

Downstream Classifier

`[I, I·R] vs [I]:` +3.11 pp overall gain
Road +11.6% • Sidewalk +7.6%





Conclusion

It Works

$\hat{\rho} = I \cdot R$ provides measurable, scene-dependent semantic gains: +14.45% on a 600-frame run.

Proxy Matters

$\log(1 + I \cdot R)$ is the robust choice for multi-window use. Direct $I \cdot R$ has explicit failure modes.

Class-Dependent

Structured surfaces (road, sidewalk) benefit most. Heterogeneous classes (vegetation) can lose.

github.com/aymisxx/reflect-aug-seg